



welcome to

applications of artificial intelligence

prof. dr. nicolas meseth

a brief history of AI

when did AI begin?

image: ai-generated (gpt-image-2.5-flare)



- 1 what do you know?
- 2 what your peers knew
- 3 let's compile the knowledge
- 4 what's next

part 1

what do you know?

what do you know about the history of AI?

one card = one milestone
the year, if you know it
block letters, big
pin it on the board

10 minutes · in pairs · there are no wrong cards



part 2

what your peers knew



last year

what your peers knew

last year's group wrote a report together: "a brief history of artificial intelligence"

part 3

let's compile the knowledge

this year

let's do it properly.

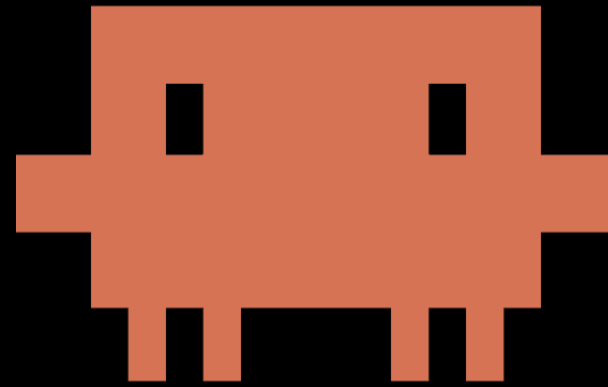
every milestone checked

every source read

every fact written down

let an AI agent help

do you know who this is?



let's ask for help.

two inputs, a photo and a text file · one task

step 1 · research

We are the master's module "Applications of Artificial Intelligence".
Help us build a knowledge base on the history of AI.

Inputs:

- a photo of our pinboard: milestones the students of this course know
- milestones.md: AI milestones that students of this course collected a year ago

Tasks:

- find reliable sources for every milestone, and research further important milestones yourself
- download every source and save it
- read each source and extract the key facts:
what? who? where? when? – and so what?
- store them in a structured knowledge base as Markdown, one file per milestone, with its source and its origin: last semester, today's students, or agent
- write all errors you find in milestones.md to an Excel list
- pinboard cards you cannot verify go to a list "still to check"

First read everything and show me your plan. Wait for my go.

libraries are boring,
so let's build
something
interactive.



image: ai-generated (gpt-image-2.5-flare)

let's compile the knowledge

step 2 · build

Now build a visually compelling, interactive web app "A Brief History of AI" for the students of this module. It guides visitors through the history of AI on a timeline.

- use only the knowledge base – no new facts
- every entry shows what, who, where, when and so what, and links to its source
- show where each entry came from: last semester, today's students, agent
- a static site (HTML, CSS, JavaScript, no build step), in English
- must work on phones and on the projector; do not include the photo

Show me your plan first. Wait for my go.

while the agent builds

Why AI Agents are either the best or worst thing we've ever built

Hannah Fry



Ansehen auf

video: hannah fry, why AI agents are either the best or worst thing we've ever built (youtube)

let's compile the knowledge

test it

trust, but check.

which errors did the agent find in last year's list?

which of our cards made it, which are “still to check”?

what did the agent add on its own?

click a source: does it say what the app says?

time for
assessment.

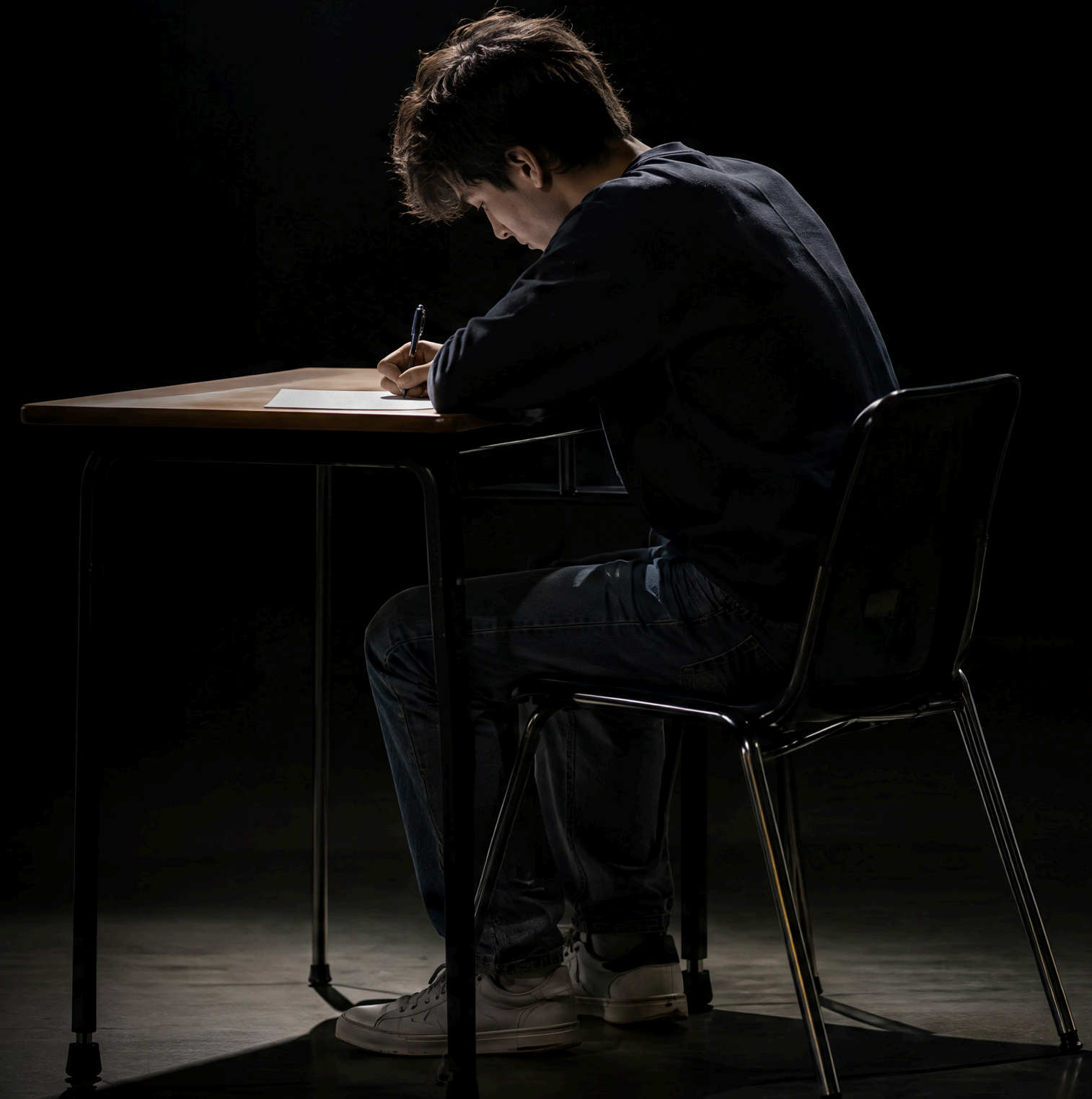


image: ai-generated (gpt-image-2.5-flare)

let's compile the knowledge

step 3 · quiz

Add a quiz to the app, so visitors can test what they learned about the history of AI.

- 50 interesting questions, based only on the knowledge base
- 4 answer options per question, exactly one of them correct
- the wrong options must be plausible, not silly
- after each answer: right or wrong, the correct answer, and a link to the matching entry on the timeline
- questions in random order, a score at the end
- same design as the timeline; must work on phones

Show me three example questions first. Wait for my go.

bonus · guided mode

Add a guided mode to the app: a narrator tells the history of AI, and the app follows along.

- write an engaging story from the milestones, based only on the knowledge base
- choose a fitting narrator voice from the OpenAI text-to-speech API; the API key is in the environment variable OPENAI_API_KEY
- create the audio in advance – the app itself never calls the API
- while the narrator speaks, the app opens the matching milestone on the timeline by itself, no clicks needed
- the user can pause, skip and leave the guided mode at any time
- say in the app that the voice is AI-generated
- never write the API key into the app or into any file

Show me the outline of the story and the voice you chose first.
Wait for my go.

publish

Create a public GitHub repository "brief-history-of-ai" and publish the app with GitHub Pages. Give me the URL.

Publish only the app, its audio and the knowledge base – not the downloaded sources, not the Excel list, not the photo.

Then make a QR code for the URL and open it in the browser, so everybody in the room can scan it.

part 4

what's next

what the agent did today, and what we'll unpack

read a photo of handwriting

→ how computers see images

what the agent did today, and what we'll unpack

read a photo of handwriting

→ how computers see images

wrote a plan

→ reasoning

what the agent did today, and what we'll unpack

read a photo of handwriting

→ how computers see images

wrote a plan

→ reasoning

generated text and code

→ how LLMs & neural networks work

what the agent did today, and what we'll unpack

read a photo of handwriting

→ how computers see images

wrote a plan

→ reasoning

generated text and code

→ how LLMs & neural networks work

knew how to build a web app

→ how an LLM is trained

what the agent did today, and what we'll unpack

read a photo of handwriting

→ how computers see images

wrote a plan

→ reasoning

generated text and code

→ how LLMs & neural networks work

knew how to build a web app

→ how an LLM is trained

used tools: files, browser, web

→ rules vs. learning · tool use

what the agent did today, and what we'll unpack

read a photo of handwriting

→ how computers see images

wrote a plan

→ reasoning

generated text and code

→ how LLMs & neural networks work

knew how to build a web app

→ how an LLM is trained

used tools: files, browser, web

→ rules vs. learning · tool use

got something wrong

→ hallucination

what the agent did today, and what we'll unpack

read a photo of handwriting

→ how computers see images

wrote a plan

→ reasoning

generated text and code

→ how LLMs & neural networks work

knew how to build a web app

→ how an LLM is trained

used tools: files, browser, web

→ rules vs. learning · tool use

got something wrong

→ hallucination

checked facts against sources

→ trust and verification

what the agent did today, and what we'll unpack

read a photo of handwriting

→ how computers see images

wrote a plan

→ reasoning

generated text and code

→ how LLMs & neural networks work

knew how to build a web app

→ how an LLM is trained

used tools: files, browser, web

→ rules vs. learning · tool use

got something wrong

→ hallucination

checked facts against sources

→ trust and verification

split the work among sub-agents

→ multi-agent systems

what the agent did today, and what we'll unpack

read a photo of handwriting

→ how computers see images

wrote a plan

→ reasoning

generated text and code

→ how LLMs & neural networks work

knew how to build a web app

→ how an LLM is trained

used tools: files, browser, web

→ rules vs. learning · tool use

got something wrong

→ hallucination

checked facts against sources

→ trust and verification

split the work among sub-agents

→ multi-agent systems

published to the web on its own

→ autonomy, safety, responsibility

how do these systems work?

what's the difference between ChatGPT and Claude Code?

how does a language model represent text, and meaning?

why does it give a different answer every time?

how much can it keep in mind at once?

can I run an AI on my own laptop?

why do the models keep getting better?

what does it mean for us?

where is AI already at work in our domains?

the machine predicts, who decides?

how should we use AI in our studies and in research?

how much energy does an answer cost?

which jobs will change?

what does AI mean for the economy?

whose data is it?

how do we regulate AI, and where is all this going?

your homework

think about whether you want answers
to these questions.

then come back next week.

📅 next week

more details for this course.



image: ai-generated (gpt-image-2.5-flare)

what's next

the next milestone
is still blank.